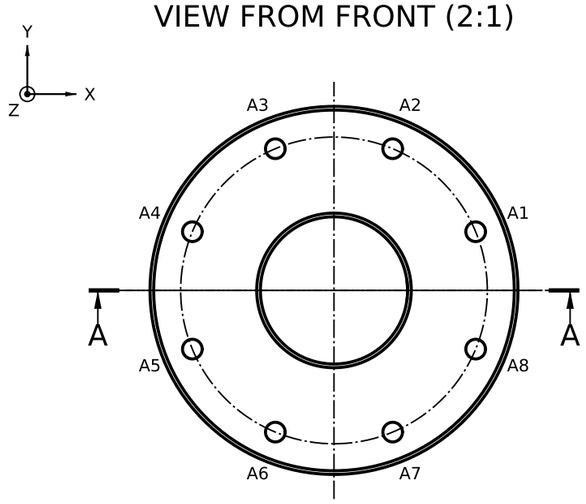
## NOTES

1. Dimensions in mm. General tolerances ISO 2768-mK. Fits ISO 286-1/-2 (deviations shown in brackets).
2. STEP file governs undimensioned geometry: cnc/step/encoder_hub.step (STEVE 02e31a10 rev 49).
3. Edges ISO 13715 -0.2 (break 0.1-0.2 x 45°) unless a chamfer is shown. Remove all burrs, especially in bores and holes.
4. Surface roughness Ra 3.2 µm unless noted (ISO 21920-1).
5. Threads ISO metric coarse, 6H internal / 6g external; thread depths are full-thread depths.
6. Material AISI 303 (1.4305): non-magnetic, keep magnetic steel away from the ring's magnet. Passivate ASTM A967. (Model shows 304.)
7. Datum G: Ø12 H7 bore; datum H: ring mounting face (flange rear face, carries the RLS MRA039 ring).
8. Bore goes on the shaft's Ø12 g6 seat (AM-09 Z1) with Loctite 638. Fit with a 0.50 mm shim against the end bell rear face while it cures (sets the 2.88 ±0.13 ring-to-readhead gap).
9. Collar Ø18 must pass the readhead's Ø21.2 bore.

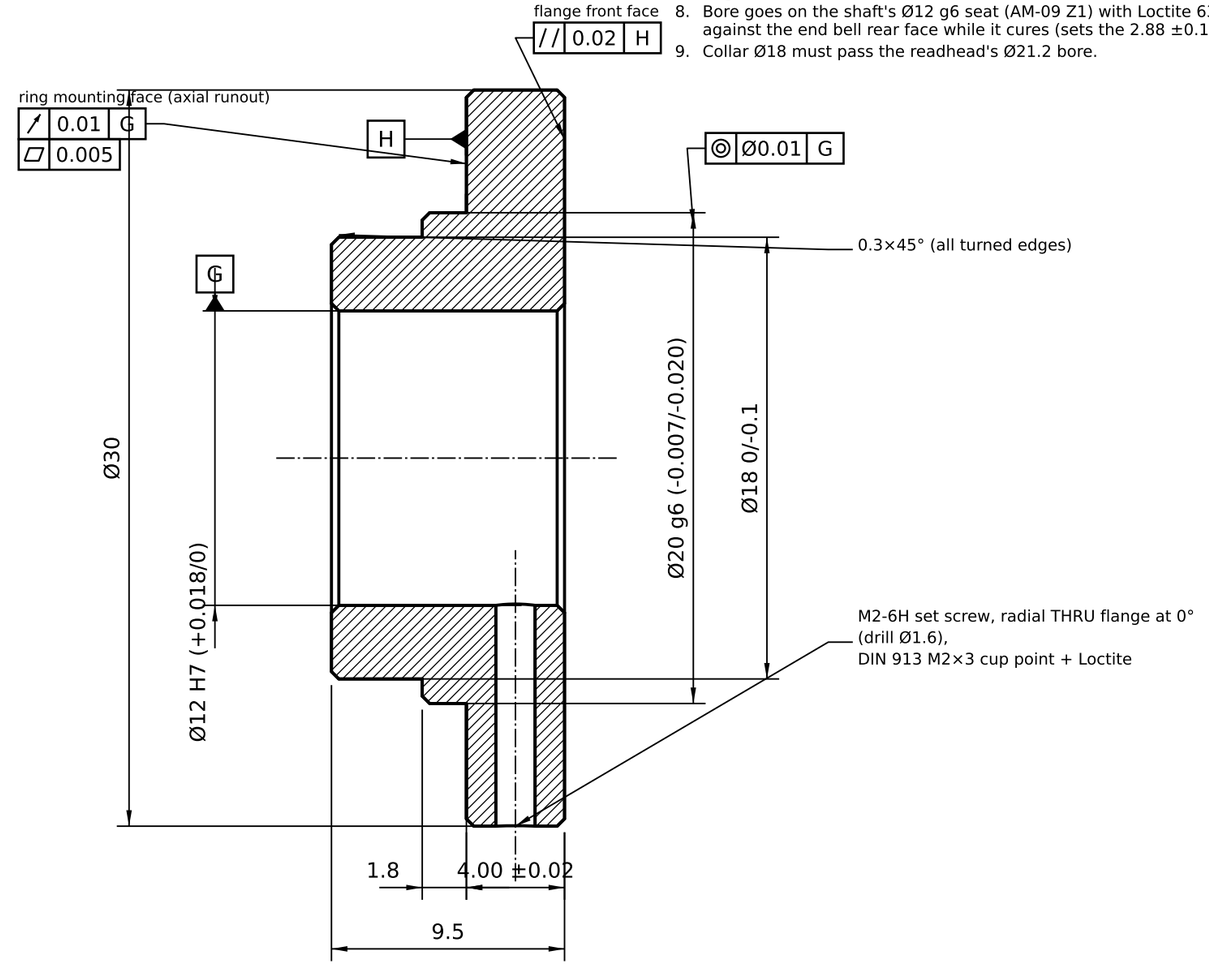
## VIEW FROM FRONT (2:1)




| TAG | X | Y | HOLE |
|---|---|---|---|
| A1 | 11.548 | 4.784 | M2-6H THRU flange (drill Ø1.6) |
| A2 | 4.784 | 11.548 | M2-6H THRU flange (drill Ø1.6) |
| A3 | -4.784 | 11.548 | M2-6H THRU flange (drill Ø1.6) |
| A4 | -11.548 | 4.784 | M2-6H THRU flange (drill Ø1.6) |
| A5 | -11.548 | -4.784 | M2-6H THRU flange (drill Ø1.6) |
| A6 | -4.784 | -11.548 | M2-6H THRU flange (drill Ø1.6) |
| A7 | 4.784 | -11.548 | M2-6H THRU flange (drill Ø1.6) |
| A8 | 11.548 | -4.784 | M2-6H THRU flange (drill Ø1.6) |

1. X/Y in the STEP frame (Z axis = datum G).
2. A1-A8: position Ø0.05 to G. ISO 7046 M2×6 ring screws, 4 engaged.

## SECTION A-A (4:1)




| REV | DESCRIPTION | DATE | BY |
|---|---|---|---|
| A | First release for quotation and manufacture | 2026-10-14 | Claude |

| | | | | | |
|---|---|---|---|---|---|
| PROJECT: **Unknown machine actuator** | DRAWN BY: Claude for macncrash | | | SOURCE (MODEL GOVERNS): STEVE 02e31a10 rev 49<br>STEP: cnc/step/encoder_hub.step | |
| GENERAL TOLERANCES: ISO 2768-mK (ISO 22081)<br>Edges ISO 13715 -0.2 (0.1-0.2) | | PROJECTION: FIRST ANGLE<br>ISO 128 / ISO 5456-2 | | | |
| FINISH / HEAT TREATMENT: Passivate ASTM A967 | | | | UNITS: mm | |
| MATERIAL: AISI 303 (1.4305) | | | | SCALE: 4:1 (views as noted) | |
| TITLE: **Encoder hub** | | | | QTY PER ACTUATOR: **1** | |
| DRAWING / PART NUMBER: **AM-11-encoder_hub** | REV: **A** | SHEET: 1/1 | SIZE: A3 | DATE OF ISSUE: 2026-10-14 | |